Cantera vs NN | Test IC 68 | NRMSE (util.py)

T0 = 1086.3 K P0 = 3.702 atm phi0 = 1.168 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
2 P-only DeepONet10	**10k**	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	**long**	0.248	0.947	0.947	9.35	0.916	2.28	0.352	1.23	1.07	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.0874	0.309	0.756	0.913	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0768	0.403	0.258	0.249	0.623	0.15	0.127	1.26	1.15	0	Complete
4 T/P DeepONet10	**10k**	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	**long**	0.48	0.362	11.2	0.253	0.283	5.35	0.284	0.924	1.07	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.3	0.302	3.93	0.68	0.431	5.48	0.291	2.17	1.34	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.296	0.376	4.5	0.345	0.578	4.66	0.273	1.91	1.16	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.236	0.366	3.17	0.598	0.563	4.14	0.237	1.49	1.18	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.